

UART frame

Asynchronous serial UART sends one byte as a timed bit sequence, no shared clock wire on the link

Starter 0xA5 as 8N1

- Starter: byte hex A5 as eight-N-one, eight data bits, no parity, one stop bit
- Idle high, start zero, then D0 through D7 LSB first, then stop one, then idle again
- For A5 the LSB is one
- Step bit by bit or play to end to watch the cursor move through twelve slots
- Rebuild after changing width, parity, or stop count

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / UART frame

INTERACTIVE TOOL

UART frame animator

Step a classic async serial frame on TX: idle-high, start=0, data LSB-first, optional parity, stop=1. Starter: byte 0xA5 in **8N1**.

Starter example: byte 0xA5 as **8N1** (8 data, no parity, 1 stop). Idle-high → start 0 → LSB-first data → stop 1.

Load starter example

Challenges 0 / 22 cleared

Quiz: idle: Between UART frames the TX line typically sits at...

☐ logic 1 (idle / mark) ☐ logic 0 always ☐ high-Z ☐ toggling every baud

Show hint

Check

Next

Idle

Quiz: idle

Quiz: start

Quiz: bit order

Quiz: 8N1

Starter

Step once

Land on start

Data D0

Stop bit

Play to end

Byte 0xA5

Even parity

Parity of A5 even

Odd parity

Two stop bits

7 data bits

Demo

Explain

8N1 length

Reset cursor

MSB on wire

Sketch config

Core ideas

Idle

Line rests at logic 1 between frames.

Start

One bit time of 0 marks the frame edge.

Data

LSB first; width often 5–9 bits.

Stop

One (or two) bit time(s) of 1; optional parity before stop.

UART frame starter

Real RTL/TB practice

- In Track A, sketch the same frame on paper
- Write what eight-N-one means in words
- Optional: peek at UART examples in the legacy combined materials and name one signal you
- This lab is frame literacy, not a full synthesizable UART yet

Pitfalls to watch

- Do not send MSB first on the wire
- Do not confuse baud with the system clock
- Parity changes the slot count; eight-E-one is not eight-N-one
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, step through starter A5 eight-N-one and land on start, D0, and stop
- On paper, draw one complete frame with bit values labeled
- When you are ready, take the short quiz, then continue to spec-to-RTL checklist